BEGINNER TO INTERMEDIATE PROGRAM

C / C++ PROGRAMMING

Pointers · OOP · Memory Management · Data Structures · File Handling — master the foundation of all modern software with C and C++.

2 Courses	22 Modules	Projects	Live Sessions
C + C++	Complete Path	Per Course	Expert-Led

YOUR LEARNING JOURNEY

Start with C fundamentals, then level up to object-oriented C++.

	M1–2	M3–4	M5–6	M7	M8	M9–11
C	Basics & Ops Variables, I/O, Operators	Control Flow Conditions & Loops	Arrays & Strings Data Collections	Functions Modular Coding	Pointers Memory & References	Structs Files & Project Advanced + Project

	M1–2	M3–4	M5–6	M7–8	M9–10	M11
C++	C++ Intro & OOP Basics Classes & Objects	Encap. & Functions Access Control	Constructors & Operators Object Lifecycle	Inheritance & Polymorphism Code Reuse	Abstraction & Relationships Design Patterns	Templates & Practice Generic Code

COURSE 1 — C PROGRAMMING

The foundation of all programming — structured, logic-driven, and close to the hardware.

MOD 01

Introduction to Programming & C Basics

- What is Programming? History & Features of C
- Installing Compiler / IDE (GCC, CodeBlocks)
- Structure of a C Program
- Keywords, Identifiers & Variables
- Data Types: int, float, char, double
- Input & Output: printf, scanf
- Type Casting & sizeof Operator
- Arithmetic & Relational Operators
- Logical & Assignment Operators
- Operator Precedence & Expressions

OUTCOME ***Understand how a C program runs and write your first programs from scratch***

MOD 02

Decision Making & Loops

- if, if-else & Nested if-else
- switch Statement
- Conditional (Ternary) Operator
- for Loop
- while Loop
- do-while Loop
- Nested Loops
- Pattern Programs (stars, numbers)
- break, continue & goto
- Loop-based logic building exercises

OUTCOME ***Write programs with full decision logic and loop-based repetition control***

MOD 03

Arrays & Strings

- Introduction to Arrays
- One-Dimensional Arrays & Programs
- Two-Dimensional Arrays (2D Matrix)
- Passing Arrays to Functions
- Introduction to Strings
- String Input / Output
- String Handling Functions (strlen, strcpy, strcat, strcmp)
- String Programs
- Passing Strings to Functions
- Character arrays vs string literals

OUTCOME ***Handle collections of data and text confidently with arrays and strings***

MOD 04

Functions, Pointers & Dynamic Memory

- What is a Function? Library vs User-Defined
- Function Declaration & Definition
- Call by Value vs Call by Reference
- Recursion — basic programs
- Pointer Introduction & Arithmetic
- Pointers with Arrays & Strings
- Dynamic Memory Allocation (malloc, calloc, free)
- Pointer to Pointer
- Function Pointers
- Common pointer pitfalls & debugging

OUTCOME ***Write modular reusable code and understand how memory works under the hood***

MOD 05

Structures, File Handling & Projects

- Structure Introduction & Declaration
- Array of Structures
- Pointer to Structure
- typedef and enum
- Storage Classes: auto, static, extern, register
- File Handling Basics — fopen, fclose
- File Read / Write / Append Operations
- File Handling Programs
- Mini Project: Logic-based application
- Final Project + Interview Questions

OUTCOME ***Model real-world data, work with files, and build complete C applications***

COURSE 2 — C++ PROGRAMMING

Level up from C to Object-Oriented C++ — classes, inheritance, polymorphism, and templates.

MOD 01

Introduction to C++ & Core Concepts

- Why C++? (C vs C++ differences)
- Basic structure of a C++ program
- iostream, cin, cout
- Namespace (std)
- Coding style & naming conventions
- Data types, variables & constants
- References (&)
- new and delete operators
- Difference: struct (C) vs class (C++)
- Basic I/O programs in C++

OUTCOME ***Transition smoothly from C to C++ and write your first object-oriented programs***

MOD 02

Classes, Objects & Encapsulation

- What is a Class & Object?
- Object Properties (Data Members) & Behavior (Methods)
- Instance variables & instance methods
- Status of object — flow & block concept
- Access modifiers: public, private, protected
- Encapsulation with real-life examples
- Instance variable hiding & this pointer
- Function definition inside & outside class
- Passing object to a function
- Inline functions

OUTCOME ***Think in objects — design clean, encapsulated classes that model real-world entities***

MOD 03

Constructors, Destructors & Operator Overloading

- Constructor & Destructor
- Types of constructors (default, parameterized, copy)
- Internal working of constructor
- Dynamic object creation
- Function overloading
- Operator overloading — introduction
- Unary operator overloading
- Binary operator overloading
- Constructor with dynamic memory
- Deep copy vs shallow copy

OUTCOME ***Control object lifecycle and enable natural operator usage with custom classes***

MOD 04

Inheritance & Polymorphism

- What is Inheritance (IS-A relationship)
- Access specifiers in inheritance
- Types: single, multilevel, multiple, hierarchical, hybrid
- Diamond problem & virtual base class
- Virtual function
- Pure virtual function
- Abstract class
- Runtime polymorphism
- Function overriding
- vtable & vptr concepts

OUTCOME ***Promote code reuse, build hierarchies, and achieve dynamic runtime behavior***

MOD 05

Object Relationships & Templates

- Has-A relationship (Composition)
- Tightly coupled vs loosely coupled objects
- Friend function & friend class
- Function templates
- Class templates
- STL basics — vector, list, map
- Template specialization
- Exception handling: try, catch, throw
- Mini Project: OOP-based application
- Final Project + Interview Questions

OUTCOME ***Write generic reusable code and design flexible, maintainable software systems***

TOOLS & TECHNOLOGIES YOU WILL USE

GCC Compiler C/C++ Build	**CodeBlocks** IDE	**VS Code** Editor	**GDB** Debugger	**Valgrind** Memory Check
Make / CMake Build Tools	**Git / GitHub** Version Control	**Linux Terminal** CLI Tools	**STL** C++ Library	**C++17** Modern C++

WHO IS THIS FOR?

- **Fresh Graduates**
  BCA, B.Tech, BSc, Diploma — any stream
- **Complete Beginners**
  Zero programming background? Start here
- **CS / IT Students**
  Strengthen fundamentals for exams & placements
- **Competitive Coders**
  Build the logic foundation for DSA & CP
- **Working Professionals**
  Add systems programming to your skillset

WHY CODING SHARKS?

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